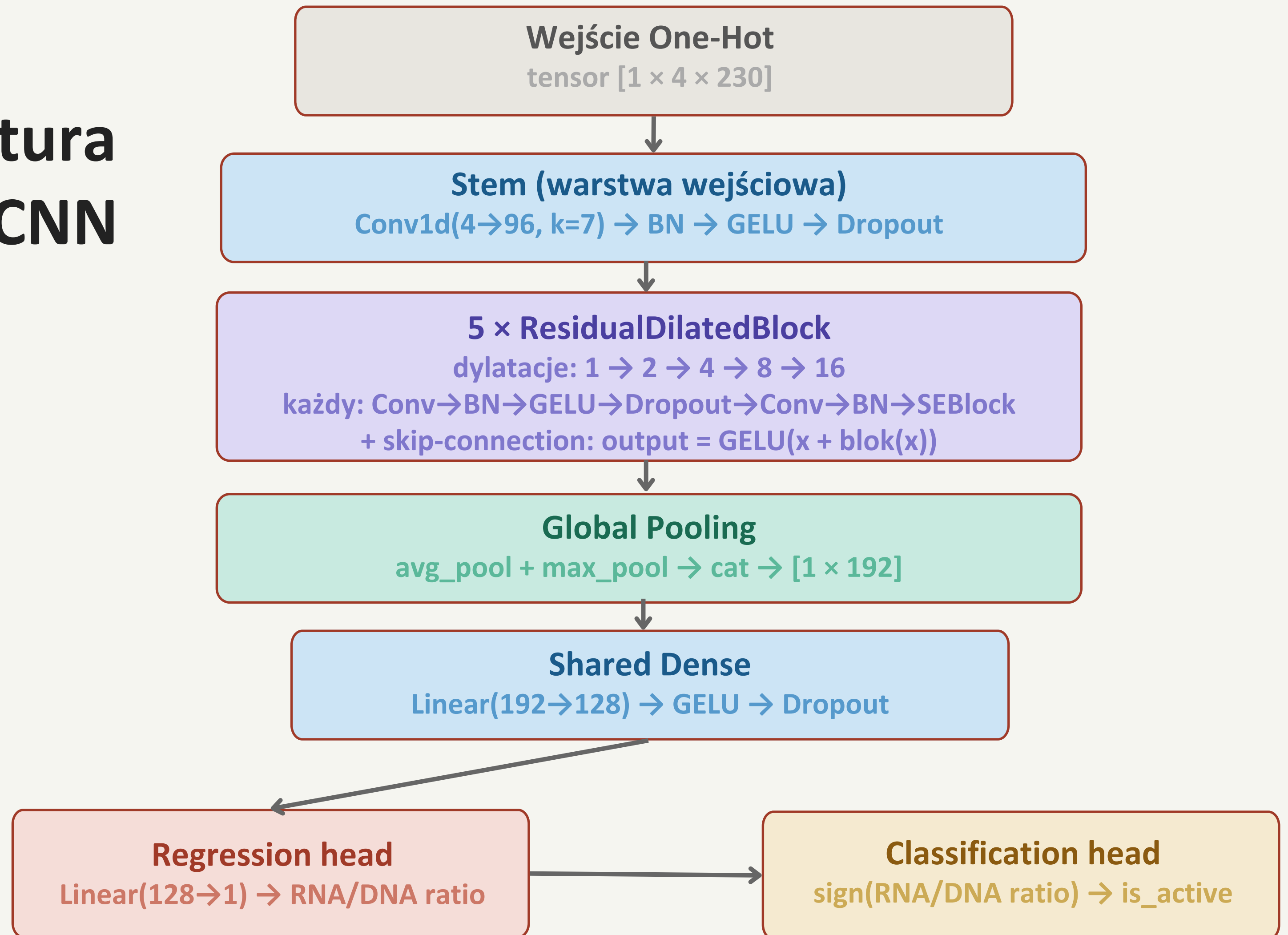


Final Group Project:

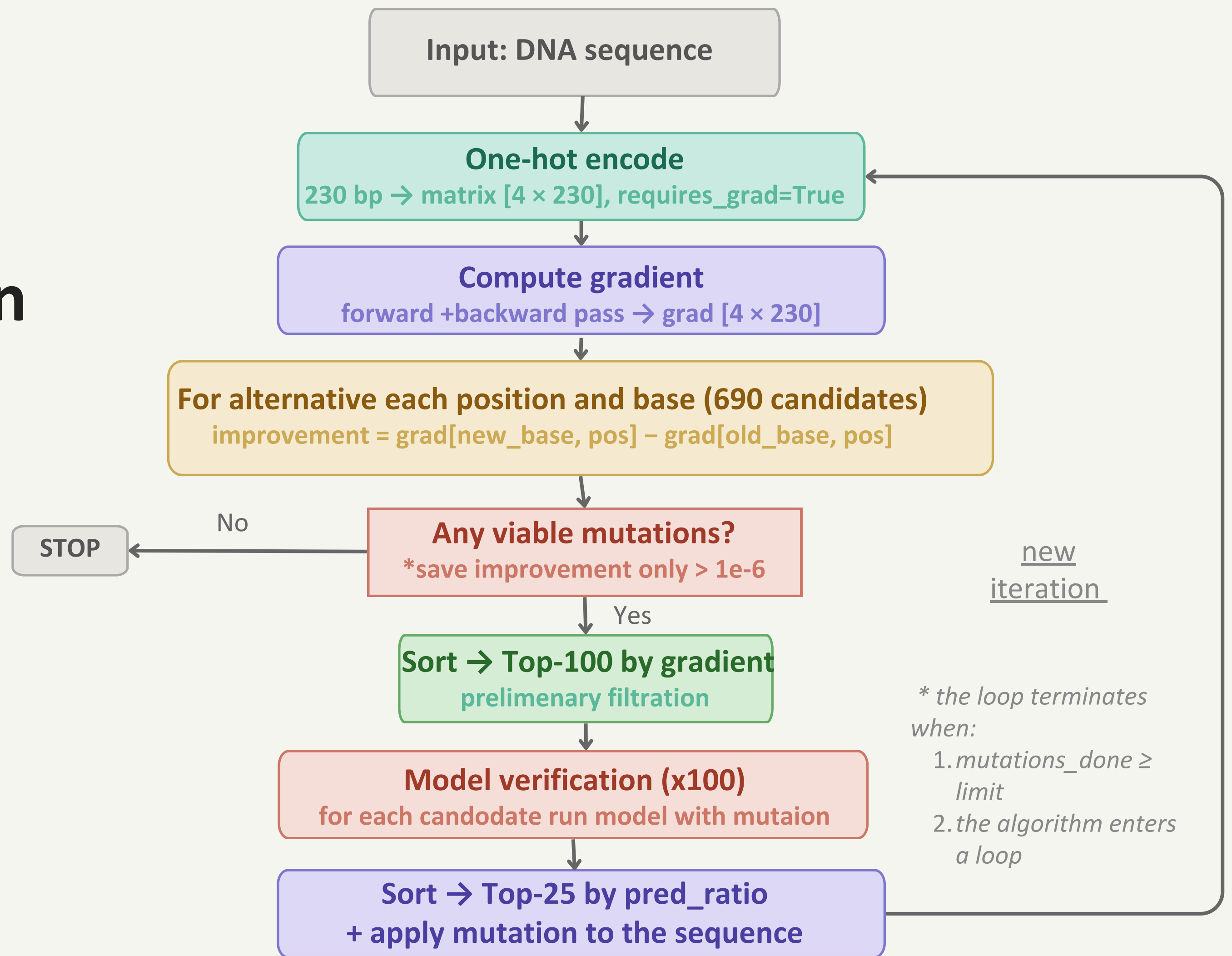
Deep Learning for Regulatory DNA Sequence Optimization

Michał Zgieb
Maciej Kowalski
Katsiaryna Dubrouskaya

Architektura modelu CNN



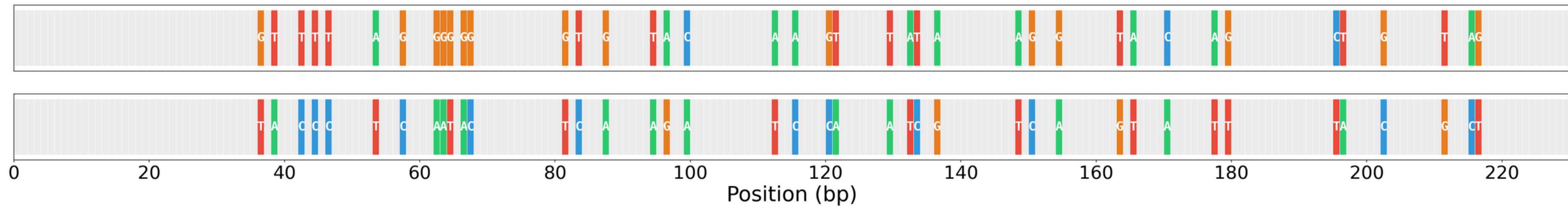
Gradient Attribution



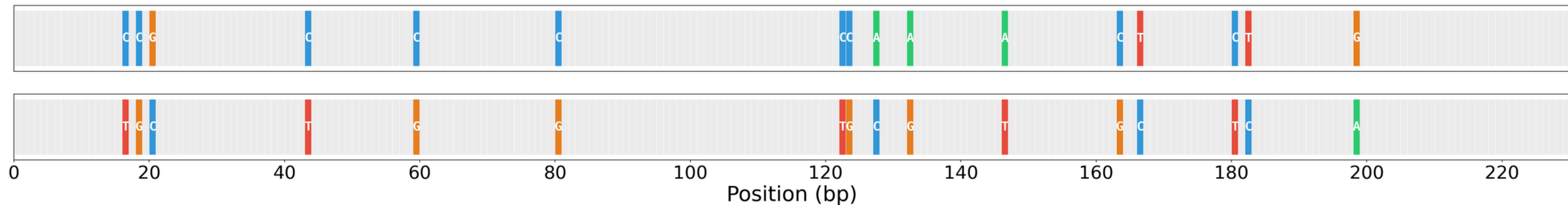
Results

Sequence	Activity before mutation	Original activity (approx.)	Introduced mutations	Achieved activity
A. Seq 1	1.38	~2.5	40	7.51
A. Seq 2	0.79	~2.7	16	6.61
A. Seq 3	0.48	~1.1	20	4.71
B. Seq	-0.48	---	80	8.77

 A  C  G  T  unchanged



 A C G T unchanged



 A C G T unchanged

